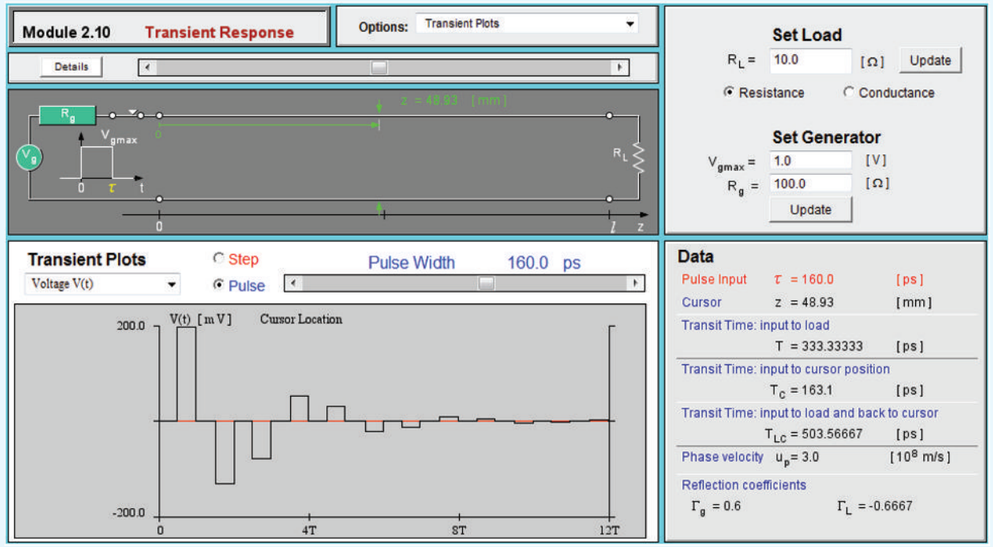
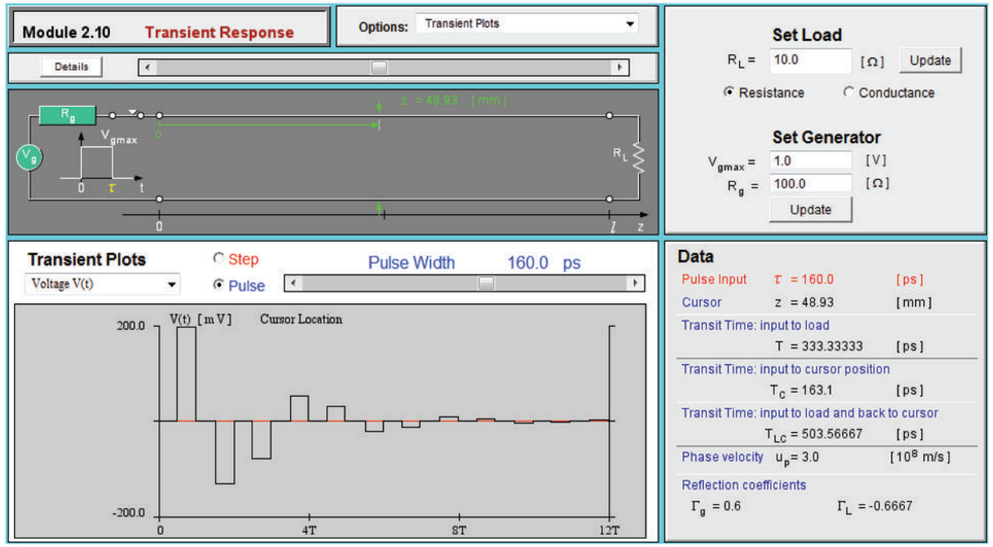
Module 2.10 Transient Response For a lossless line terminated in a resistive load, the module simulates the dynamic response—at any location on the line—to either a step or pulse waveform sent by the generator.



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模块 2.10 瞬态响应 对于端接电阻负载的无损线路，该模块模拟线路上任何位置对发生器发送的阶跃波形或脉冲波形的动态响应。



Technology Brief 4: EM Cancer Zappers

From laser eye surgery to 3-D X-ray imaging, EM sources and sensors have been used as medical diagnostic and treatment tools for many decades. Future advances in information processing and other relevant technologies will undoubtedly lead to the greater performance and utility of EM devices, as well as to the introduction of entirely new types of devices. This Technology Brief introduces two recent EM technologies that are still in their infancy but are quickly developing into serious techniques for the surgical treatment of cancer tumors.

Microwave Ablation

In medicine, ***ablation*** is defined as the "surgical removal of body tissue," usually through the direct application of chemical or thermal therapies.

► Microwave ablation applies the same heat-conversion process used in a microwave oven (see TB3), but instead of using microwave energy to cook food, it is used instead to destroy cancerous tumors by exposing them to a focused beam of microwaves. ◄

The technique can be used ***percutaneously*** (through the skin), ***laparoscopically*** (via an incision), or ***intraoperatively*** (open surgical access). Guided by an imaging system, such as a CT scanner or an ultrasound imager, the surgeon can localize the tumor and then insert a thin coaxial transmission line (∼1.5 mm in diameter) directly through the body to position the tip of the transmission line (a probe-like antenna) inside the tumor (**Fig. TF4-1**). The transmission line is connected to a generator capable of delivering 60 W of power at 915 MHz (**Fig. TF4-2**). The rise in temperature of the tumor is related to the amount of microwave energy it receives, which is equal to the product of the generator's power level and the duration of the ablation treatment. Microwave ablation is a promising new technique for the treatment of liver, lung, and adrenal tumors.

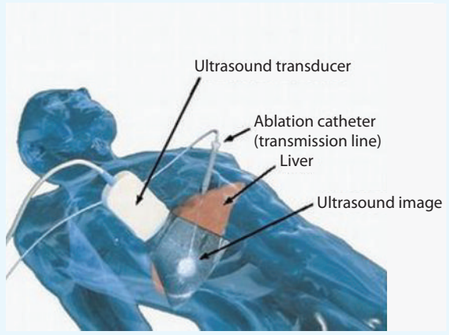


Figure TF4-1 Microwave ablation for liver cancer treatment.

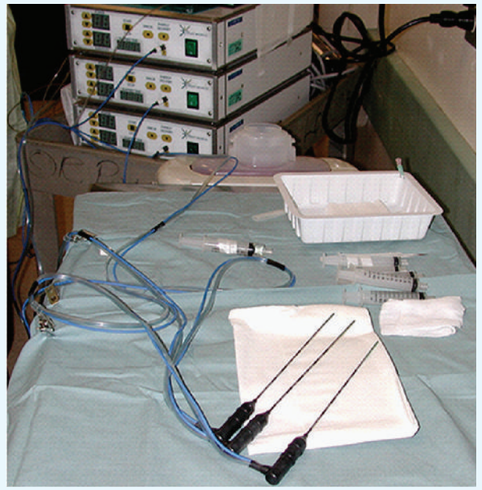


Figure TF4-2 Photograph of the setup for a percutaneous microwave ablation procedure in which three single microwave applicators are connected to three microwave generators. (Courtesy of *RadioGraphics*, October 2005, pp. 569–583.)

High-Power Nanosecond Pulses

Bioelectrics is an emerging field focused on the study of how electric fields behave in biological systems. Of particular recent interest is the desire to understand how living cells might respond to the application of extremely short pulses (on the order of nanoseconds (10^{-9} s) and even as short as picoseconds (10^{-12} s)) with exceptionally high voltage and current amplitudes.

技术简介 4：电磁癌症消灭器

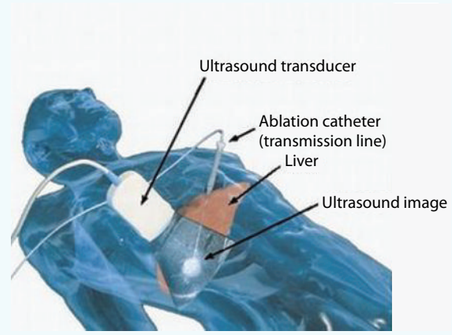
从激光眼科手术到 3D X 射线成像，数十年来，电磁源和传感器一直被用作医疗诊断和治疗工具。信息处理和其他相关技术的未来进步无疑将提高电磁设备的性能和实用性，并引入全新类型的设备。本技术简介介绍了两项最新的 EM 技术，它们仍处于起步阶段，但正在迅速发展成为用于癌症肿瘤外科治疗的重要技术。

微波消融

在医学上，消融被定义为“手术切除身体组织”，通常通过直接应用化学或热疗法。

► 微波消融采用与微波炉相同的热转换过程（见TB 3），但不是使用微波能量来烹饪食物，而是通过将癌性肿瘤暴露在聚焦的微波束下来摧毁它们。 ◄

该技术可经皮（通过皮肤）、腹腔镜（通过切口）或术中（开放手术通道）使用。在 CT 扫描仪或超声成像仪等成像系统的引导下，外科医生可以定位肿瘤，然后插入细同轴传输线（∼直径 1.5 毫米）直接穿过身体，将传输线（探针状天线）的尖端定位在肿瘤内部（图 TF4-1）。传输线连接到能够以 915 MHz 提供 60 W 功率的发电机（图 TF4-2）。肿瘤温度的升高与其接收的微波能量有关，该能量等于发生器的功率水平与消融治疗持续时间的乘积。微波消融是治疗肝、肺和肾上腺肿瘤的一种有前途的新技术。



图TF4-1 微波消融治疗肝癌。

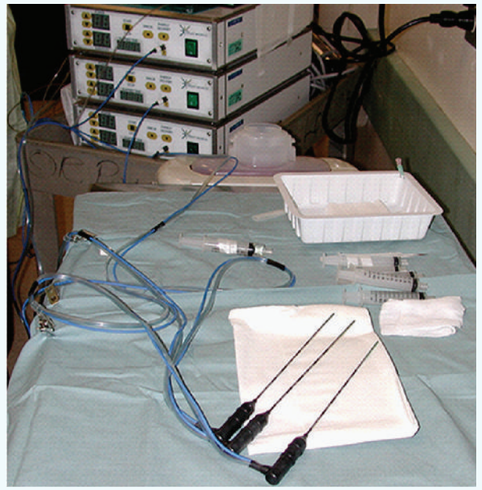


图 TF4-2 经皮微波消融手术的设置照片，其中三个单个微波发生器连接到三个微波发生器。 （RadioGraphics 提供，2005 年 10 月，第 569-583 页。）

高功率纳秒脉冲

生物电学是一个新兴领域，专注于研究电场在生物系统中的行为方式。最近特别感兴趣的是了解的愿望

活细胞如何响应具有极高电压和电流幅度的极短脉冲（纳秒 (10^{-9} s) 量级，甚至短至皮秒 (10^{-12} s)）。

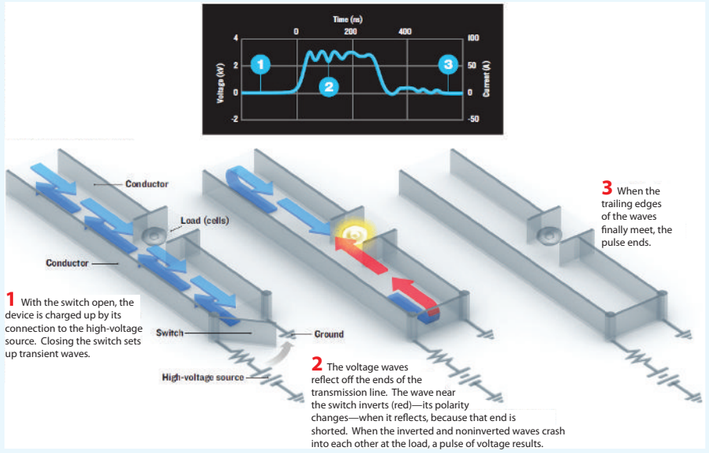


Figure TF4-3 High-voltage nanosecond pulse delivered to tumor cells via a transmission line. The cells to be shocked by the pulse sit in a break in one of the transmission-line conductors. (Courtesy of *IEEE Spectrum*, August 2006.)

► The motivation is to treat cancerous cells by ***zapping*** them with high-power pulses. The pulse power is delivered to the cell via a transmission line, as illustrated by the example in **Fig. TF4-3**. ◄

Note that the pulse is about 200 ns long, and its voltage and current amplitudes are approximately 3,000 V and 60 A, respectively. Thus, the peak power level is about 180,000 W! However, the total energy carried by the pulse is only $(1.8 \times 10^5) \times (2 \times 10^{-7}) = 0.0036$ joules. Despite the low energy content, the very high voltage appears to be very effective in destroying malignant tumors (in mice, so far), with no regrowth.

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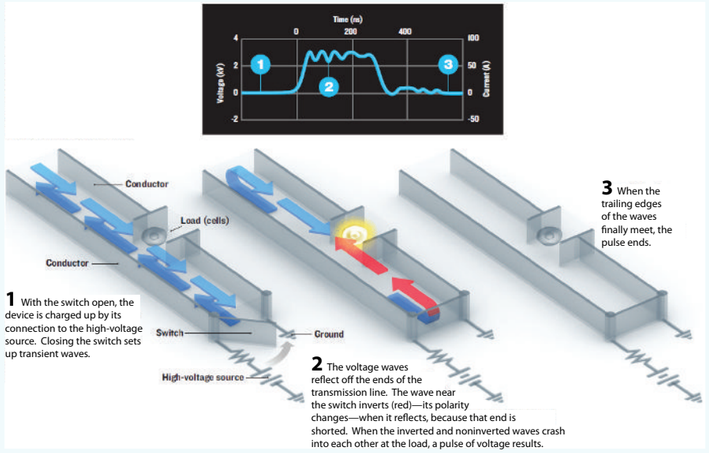


图 TF4-3 通过传输线传送至肿瘤细胞的高压纳秒脉冲。受到脉冲冲击的细胞位于一根传输线导体的断口处。（由 IEEE Spectrum 提供，2006 年 8 月。）

► 其动机是通过用高功率脉冲攻击癌细胞来治疗它们。脉冲功率通过传输线传送到电池，如图 TF4-3 中的示例所示。 ◄

请注意，脉冲长约 200 ns，其电压和电流幅度分别约为 3,000 V 和 60 A。因此，峰值功率约为 180,000 W！然而，脉冲携带的总能量仅为$(1.8\times10^5)\times(2\times10^{-7})=0.0036$焦耳。尽管能量含量较低，但非常高的电压似乎可以非常有效地摧毁恶性肿瘤（迄今为止在小鼠中），并且不会再生。

Chapter 2 Summary

Concepts

- A transmission line is a two-port network connecting a generator to a load. EM waves traveling on the line may experience ohmic power losses, dispersive effects, and reflections at the generator and load ends of the line. These transmission-line effects may be ignored if the line length is much shorter than λ.
- TEM transmission lines consist of two conductors that can support the propagation of transverse electromagnetic waves characterized by electric and magnetic fields that are transverse to the direction of propagation. TEM lines may be represented by a lumped-element model consisting of four line parameters (R', L', G', and C') whose values are specified by the specific line geometry, the constitutive parameters of the conductors and of the insulating material between them, and the angular frequency ω.
- Wave propagation on a transmission line, which is represented by the phasor voltage $\tilde{V}(z)$ and associated current $\tilde{I}(z)$, is governed by the propagation constant of the line, $\gamma = \alpha + j\beta$, and its characteristic impedance Z_0. Both γ and Z_0 are specified by ω and the four line parameters.
- If $R' = G' = 0$, the line becomes lossless ($\alpha = 0$). A lossless line is generally nondispersive, meaning that the phase velocity of a wave is independent of the frequency.
- In general, a line supports two waves, an incident wave supplied by the generator and another wave reflected by the load. The sum of the two waves generates a standing-wave pattern with a period of $\lambda/2$. The voltage standing-wave ratio S, which is equal to the ratio of the maximum to minimum voltage magnitude on the line, varies between 1 for a matched load ($Z_L = Z_0$) to ∞ for a line terminated in an open circuit, a short circuit, or a purely reactive load.
- The input impedance of a line terminated in a short circuit or open circuit is purely reactive. This property can be used to design equivalent inductors and capacitors.
- The fraction of the incident power delivered to the load by a lossless line is equal to $(1 - |\Gamma|^2)$.
- The Smith chart is a useful graphical tool for analyzing transmission-line problems and for designing impedance-matching networks.
- Matching networks are placed between the load and the feed transmission line for the purpose of eliminating reflections toward the generator. A matching network may consist of lumped elements in the form of capacitors and/or inductors, or it may consist of sections of transmission lines with appropriate lengths and terminations.
- Transient analysis of pulses on transmission lines can be performed using a bounce-diagram graphical technique that tracks reflections at both the load and generator ends of the transmission line.

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Chapter 2 Summary

概念

- 传输线是将发电机连接到负载的二端口网络。在线路上传播的电磁波可能会经历欧姆功率损耗、色散效应以及线路发电机和负载端的反射。如果线路长度比 λ 短得多，则可以忽略这些传输线效应。
- TEM 传输线由两个导体组成，可以支持横向电磁波的传播，其特征是电场和磁场垂直于传播方向。TEM 线可以用集总元件模型表示，该模型由四个线参数（R′、L′、G′ 和 C′）组成，其值由特定的线几何形状、导体及其之间的绝缘材料的本构参数以及角频率 ω 指定。
- 传输线上的波传播由相量电压 $\tilde{V}(z)$ 和相关电流 $\tilde{I}(z)$ 表示，由传输线的传播常数 $\gamma = \alpha + j\beta$ 及其特征阻抗 Z_0 控制。 γ 和 Z_0 均由 ω 和四个行参数指定。
- 如果 $R' = G' = 0$，则线路变为无损 ($\alpha = 0$)。无损线通常是非色散的，这意味着波的相速度与频率无关。
- 一般来说，一条线支持两种波，一种是由发生器提供的入射波，另一种是反射波由负载。两个波的总和产生周期为λ/2的驻波图案。电压驻波比 S 等于线路上最大与最小电压幅度之比，在匹配负载 ($Z_L = Z_0$) 的 1 到开路、短路或纯无功负载端接的线路的 ∞ 之间变化。
- 短路或开路端接线路的输入阻抗是纯电抗性的。该特性可用于设计等效电感器和电容器。
- 通过无损线路传送到负载的入射功率的部分等于 $(1 - |\Gamma|^2)$。
- 史密斯圆图是分析传输线问题和设计阻抗匹配网络的有用图形工具。
- 匹配网络放置在负载和馈电传输线之间，目的是消除对发电机的反射。匹配网络可以由电容器和/或电感器形式的集总元件组成，或者它可以由具有适当长度和终端的传输线部分组成。
- 传输线上的脉冲瞬态分析可以使用反弹图图形技术来执行，该技术跟踪传输线负载端和发电机端的反射。

Mathematical and Physical Models

TEM Transmission Lines

$$L'C' = \mu\varepsilon$$

$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon}$$

$$\alpha = \Re\mathrm{e}(\gamma) = \Re\mathrm{e}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(Np/m)}$$

$$\beta = \Im\mathrm{m}(\gamma) = \Im\mathrm{m}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(rad/m)}$$

$$Z_0 = \frac{R'+j\omega L'}{\gamma} = \sqrt{\frac{R'+j\omega L'}{G'+j\omega C'}} \quad (\Omega)$$

$$\Gamma = \frac{z_\mathrm{L}-1}{z_\mathrm{L}+1}$$

Step Function Transient Response

$$V_1^+ = \frac{V_\mathrm{g} Z_0}{R_\mathrm{g}+Z_0}$$

$$V_\infty = \frac{V_\mathrm{g} R_\mathrm{L}}{R_\mathrm{g}+R_\mathrm{L}}$$

$$\Gamma_\mathrm{g} = \frac{R_\mathrm{g}-Z_0}{R_\mathrm{g}+Z_0}$$

$$\Gamma_\mathrm{L} = \frac{R_\mathrm{L}-Z_0}{R_\mathrm{L}+Z_0}$$

Lossless Line

$$\alpha = 0$$

$$\beta = \omega\sqrt{L'C'}$$

$$Z_0 = \sqrt{\frac{L'}{C'}}$$

$$u_\mathrm{p} = \frac{1}{\sqrt{\mu\varepsilon}} \quad \text{(m/s)}$$

$$\lambda = \frac{u_\mathrm{p}}{f} = \frac{c}{f}\frac{1}{\sqrt{\varepsilon_\mathrm{r}}} = \frac{\lambda_0}{\sqrt{\varepsilon_\mathrm{r}}}$$

$$d_{\max} = \frac{\theta_\mathrm{r}\lambda}{4\pi} + \frac{n\lambda}{2}$$

$$d_{\min} = \frac{\theta_\mathrm{r}\lambda}{4\pi} + \frac{(2n+1)\lambda}{4}$$

$$S = \frac{1+|\Gamma|}{1-|\Gamma|}$$

$$P_\mathrm{av} = \frac{|V_0^+|^2}{2Z_0}\left[1-|\Gamma|^2\right]$$

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数学与物理模型

TEM传输线

$$L'C' = \mu\varepsilon$$

$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon}$$

$$\alpha = \Re\mathrm{e}(\gamma) = \Re\mathrm{e}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(Np/m)}$$

$$\beta = \Im\mathrm{m}(\gamma) = \Im\mathrm{m}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(rad/m)}$$

$$Z_0 = \frac{R'+j\omega L'}{\gamma} = \sqrt{\frac{R'+j\omega L'}{G'+j\omega C'}} \quad (\Omega)$$

$$\Gamma = \frac{z_\mathrm{L}-1}{z_\mathrm{L}+1}$$

阶跃激励的暂态响应

$$V_1^+ = \frac{V_\mathrm{g} Z_0}{R_\mathrm{g}+Z_0}$$

$$V_\infty = \frac{V_\mathrm{g} R_\mathrm{L}}{R_\mathrm{g}+R_\mathrm{L}}$$

$$\Gamma_\mathrm{g} = \frac{R_\mathrm{g}-Z_0}{R_\mathrm{g}+Z_0}$$

$$\Gamma_\mathrm{L} = \frac{R_\mathrm{L}-Z_0}{R_\mathrm{L}+Z_0}$$

无损传输线

$$\alpha = 0$$

$$\beta = \omega\sqrt{L'C'}$$

$$Z_0 = \sqrt{\frac{L'}{C'}}$$

$$u_\mathrm{p} = \frac{1}{\sqrt{\mu\varepsilon}} \quad \text{(m/s)}$$

$$\lambda = \frac{u_\mathrm{p}}{f} = \frac{c}{f}\frac{1}{\sqrt{\varepsilon_\mathrm{r}}} = \frac{\lambda_0}{\sqrt{\varepsilon_\mathrm{r}}}$$

$$d_{\max} = \frac{\theta_\mathrm{r}\lambda}{4\pi} + \frac{n\lambda}{2}$$

$$d_{\min} = \frac{\theta_\mathrm{r}\lambda}{4\pi} + \frac{(2n+1)\lambda}{4}$$

$$S = \frac{1+|\Gamma|}{1-|\Gamma|}$$

$$P_\mathrm{av} = \frac{|V_0^+|^2}{2Z_0}\left[1-|\Gamma|^2\right]$$

使用条件：波长式 $\lambda=\lambda_0/\sqrt{\varepsilon_r}$ 假设 $\mu_r=1$。驻波极大、极小位置须取整数 n，使距离落在传输线实际范围内；Γ=0 时无驻波极大、极小之分。